GET STARTED TODAY

LUKE GINN

AZURE

CLOUD

MASTERY

Unit 17 Introduction – Processing Non-Text Inputs

AZURE CLOUD MASTERY

Unit 16 covered agent memory. Unit 17 covers multimodal integration – enabling agents to process images, audio, and documents as tools.

- **Review of Unit 16**: You learned to store agent memory. Now you need agents to see images, hear speech, and read documents – not just process text.
- **What This Unit Covers**: You will learn Vision (analyze images), Speech (transcribe audio, speak responses), and Document Intelligence (extract data from PDFs/forms).
- **Why Multimodal Matters**: Real-world user inputs include photos, voice messages, and scanned documents. Text-only agents cannot handle these.

Exam Takeaway: Multimodal Integration is part of Agentic Enhancement domain (10% weighting). The exam tests calling Vision, Speech, and Document Intelligence APIs as tools.

What Is Multimodal? – Beyond Text

AZURE CLOUD MASTERY

Multimodal means an agent can accept multiple input types (text, image, audio, document) and produce multiple output types.

- **Input Modalities**: User sends a photo of a damaged product, a voice recording of a complaint, or a PDF contract. Agent processes all three.
- **Output Modalities**: Agent can respond with text, or speak an answer using text-to-speech, or return a highlighted image.
- **Multimodal as Tools**: Your agent does not need built-in multimodal capabilities. It calls Vision, Speech, and Document Intelligence as separate tools.

Exam Takeaway: Multimodal = multiple input/output types. Agents use external Azure AI services as tools for each modality.

Azure AI Vision – Image Analysis Tool

AZURE CLOUD MASTERY

Azure AI Vision is a service that analyzes images to extract descriptions, objects, text (OCR), and adult content ratings.

- **Vision Capabilities**: Image can describe what is in the picture, detect objects and faces, read printed/handwritten text, and detect adult or violent content.
- **Use Case for Agents**: User uploads photo of a broken product. Agent calls Vision to extract damage description, then processes refund request.
- **Input Format**: Send image as URL or raw binary data (base64-encoded). Vision returns JSON with analysis results.

Exam Takeaway: Vision API extracts information from images. Agents use it as a tool to understand visual user inputs.

Optical Character Recognition (OCR) with Vision

AZURE CLOUD MASTERY

OCR (Optical Character Recognition) extracts text from images – photos of documents, screenshots, whiteboards, or street signs.

- **OCR Definition**: Vision reads text embedded in images. Extracted text includes position, line breaks, and confidence scores for each character.
- **Use Case**: User sends a photo of a receipt. Agent calls Vision to extract receipt text, then parses it for date, amount, and vendor.
- **Language Support**: Vision OCR supports printed and handwritten text in over 100 languages. Automatically detects language from image.

Exam Takeaway: OCR extracts text from images. The exam tests when to use OCR (text in images) vs. direct text input.

Coding Pattern – Calling Vision API

AZURE CLOUD MASTERY

Your agent code calls Azure AI Vision using the SDK or REST, sending an image URL or binary data, and receives JSON with analysis results.

- **SDK Pattern**: `from azure.ai.vision import ImageAnalysisClient`. `client.analyze(image_url, visual_features=[VisualFeatures.CAPTION, VisualFeatures.OCR])`.
- **REST Pattern**: POST to `https://.cognitiveservices.azure.com/vision/v4.0/analyze`. Body includes `url` or `imageData` (base64). Include `features` parameter.
- **Response Handling**: Extract `caption.text` (description), `ocrResult.lines` (text from image), `objects` (detected items). Return relevant fields to user.

Exam Takeaway: Call Vision API with image URL or base64. Parse JSON response for captions, OCR text, and objects.

Azure AI Speech – Speech-to-Text (Transcription)

AZURE CLOUD MASTERY

Azure AI Speech provides speech-to-text (STT) that converts audio input (voice messages, recordings) into text for agent processing.

- **Speech-to-Text Definition**: STT takes an audio file or stream and returns a text transcript. Supports real-time or batch transcription.
- **Use Case**: User leaves a voice message on a support line. Agent calls STT to transcribe message, then processes intent from text.
- **Language and Customization**: Supports over 100 languages. You can train custom models for domain-specific terms (product names, technical jargon).

Exam Takeaway: Speech-to-text converts audio to text. Agents use STT to process voice inputs as if they were typed text.

Azure AI Speech – Text-to-Speech (Voice Output)

Text-to-speech (TTS) converts agent response text into spoken audio, enabling voice-based interactions for phone and smart speaker agents.

- **Text-to-Speech Definition**: TTS takes text and returns an audio file (MP3, WAV) spoken by a neural voice. Supports multiple voices and languages.
- **Use Case**: Agent processes a refund request and returns "Your refund has been approved." TTS converts this to audio for a phone call.
- **SSML Control**: Speech Synthesis Markup Language (SSML) lets you control pitch, rate, and emphasis: `<prosody rate="slow">Your refund is approved</prosody>`.

Exam Takeaway: Text-to-speech converts text to audio. Use SSML for natural-sounding speech control.

Coding Pattern – Speech-to-Text with SDK

AZURE CLOUD MASTERY

Your agent code calls Azure AI Speech to transcribe audio files using the Speech SDK or REST API, receiving JSON with transcribed text and confidence.

- **SDK Pattern**: `import azure.cognitiveservices.speech as speechsdk`. Create config with key and region. `recognizer.recognize_once_async().get().text`.
- **REST Pattern**: Send audio to `https://.cognitiveservices.azure.com/speechtotext/transcriptions:transcribe`. Body includes audio binary or URL.
- **Response Handling**: Extract `DisplayText` (transcribed text) and `Confidence` (0.0 to 1.0). If confidence < 0.7, ask user to repeat or re-record.

Exam Takeaway: Use Speech SDK for simplicity or REST for control. Parse `DisplayText` and `Confidence` from response.

Coding Pattern – Text-to-Speech with SDK

AZURE CLOUD MASTERY

Your agent code calls Azure AI Speech to convert text response to audio, which can be played to the user or saved as a file.

- **SDK Pattern**: `speech_config = speechsdk.SpeechConfig(subscription=key, region=region)`. `synthesizer = speechsdk.SpeechSynthesizer(speech_config=speech_config, audio_config=None)`.
- **Synthesize**: `result = synthesizer.speak_text_async(response_text).get()`. Audio data is in `result.audio_data`. Save to file or stream.
- **SSML Usage**: `synthesizer.speak_ssml_async(ssml_string).get()` for better voice control. Build SSML string dynamically from agent response.

Exam Takeaway: Text-to-speech returns audio binary. Use SSML for prosody and voice customization.

Azure AI Document Intelligence extracts structured data from PDFs, images, and Office documents – forms, invoices, receipts, contracts.

- **Document Intelligence Definition**: Previously called Form Recognizer. Analyzes documents to extract key-value pairs, tables, and text.
- **Pre-built Models**: Use pre-built models for invoices (extract vendor, amount, due date), receipts (merchant, date, total), ID documents (name, document number), and W-2 forms.
- **Custom Models**: Train custom models for your specific document layouts (e.g., purchase orders, inspection reports). Upload 5-10 examples for training.

Exam Takeaway: Document Intelligence extracts structured data from documents. Pre-built models cover common types; custom models for unique layouts.

Document Intelligence Use Cases for Agents

AZURE CLOUD MASTERY

Agents use Document Intelligence to process user-uploaded documents, extracting data without requiring users to type information manually.

- **Receipt Processing**: User uploads a receipt photo. Agent calls Document Intelligence to extract merchant, date, items, and total. Agent then processes expense report.
- **Invoice Approval**: Supplier uploads invoice PDF. Agent extracts amount and due date, checks against purchase order, and submits for approval.
- **Identity Verification**: User uploads driver's license. Agent extracts name and document number, compares to account on file, and verifies identity.

Exam Takeaway: Document Intelligence automates data entry from documents. The exam tests selecting the correct pre-built model for document type.

Coding Pattern – Document Intelligence with SDK

Your agent code calls Document Intelligence using the SDK or REST, sending a document URL or binary data, and receives JSON with extracted fields.

- **SDK Pattern**: `from azure.ai.formrecognizer import DocumentAnalysisClient`. `client.begin_analyze_document(model_id="prebuilt-invoice", document=file_binary)`.
- **Model ID Options**: `prebuilt-invoice`, `prebuilt-receipt`, `prebuilt-idDocument`, `prebuilt-tax.us.w2`, or custom model ID for trained models.
- **Response Handling**: `result = poller.result()`. Access fields: `result.documents[0].fields["VendorName"].value`, `fields["InvoiceTotal"].value`. Each field has `confidence`.

Exam Takeaway: Document Intelligence returns JSON with extracted fields and confidence scores. Use `result.documents[0].fields` to access data.

Multimodal Tool Integration – One Agent, Multiple Modalities

A single agent can be equipped with multiple multimodal tools – Vision, Speech, Document Intelligence – and decide which to use based on user input type.

- **Tool Definition**: Define tools for `analyze_image`, `transcribe_audio`, `extract_document`. Agent reads tool descriptions and chooses based on user upload.
- **Agent Reasoning**: User says "Here is a photo of a receipt." Agent calls `analyze_image` with OCR. User says "Here is a PDF invoice." Agent calls `extract_document`.
- **System Instruction**: "If user uploads an audio file, call \`transcribe_audio\` first, then process the transcribed text." Guides tool selection.

Exam Takeaway: Agents use multimodal tools by reading file types and selecting appropriate tool. System instructions guide tool selection.

Managing API Keys for Multimodal Services

AZURE CLOUD MASTERY

Vision, Speech, and Document Intelligence are separate Azure services, each with its own endpoint URL and API key, managed securely in configuration.

- **Service Provisioning**: In Azure portal, create separate resources for Vision, Speech, and Document Intelligence. Each has unique endpoint and two keys.
- **Storing Keys**: Use environment variables: `VISION_KEY`, `VISION_ENDPOINT`, `SPEECH_KEY`, `SPEECH_REGION`, `DOC_INTEL_KEY`, `DOC_INTEL_ENDPOINT`.
- **Managed Identity**: Enable managed identity on Agent Service. Grant roles: `Cognitive Services User` for each service. No keys needed in code.

Exam Takeaway: Each multimodal service has its own keys. Use environment variables or managed identity. The exam tests secure key management across services.

Foundry Trace for Multimodal Calls

Foundry Trace records each multimodal API call (Vision, Speech, Document Intelligence) as a span, tracking latency, token usage, and any errors.

- **Vision Span**: Span name `"vision_analyze"`. Attributes: `image_size_bytes`, `features_requested` (caption, ocr), `objects_detected_count`, `duration_ms`.
- **Speech Span**: Span name `"speech_to_text"` or `"text_to_speech"`. Attributes: `audio_duration_seconds`, `transcript_confidence`, `voice_name`, `audio_size_bytes`.
- **Document Intelligence Span**: Span name `"doc_intel_analyze"`. Attributes: `model_id` (prebuilt-invoice), `pages_processed`, `fields_extracted_count`, `average_confidence`.

Exam Takeaway: Instrument multimodal calls with spans recording service type, input size, and confidence scores.

Unit 17 Summary – Multimodal Agent Enhancement

You have learned to equip agents with Vision (image analysis, OCR), Speech (STT and TTS), and Document Intelligence (form extraction) as tools.

- **Vision Tools**: Image captioning (describe scenes), OCR (extract text from images), object detection (identify items in photos). Use for photos, screenshots, documents.
- **Speech Tools**: Speech-to-text (transcribe voice inputs), Text-to-speech (speak responses). Use for voice assistants, phone systems, accessibility.
- **Document Intelligence**: Extract structured data from invoices, receipts, ID documents, forms. Use for automated data entry and document processing.
- **Coding Skills**: Call each service with SDK or REST. Parse JSON responses for extracted data, text, and confidence scores. Manage separate API keys.
- **Observability**: Trace multimodal calls with spans recording service type, input size, and confidence.
- **Next Unit Preview**: Unit 18 introduces Input Transformation – preprocessing user prompts with NLP services before routing to agent core.

Exam Takeaway: Multimodal Integration is 10% of the exam. Master Vision (OCR, captioning), Speech (STT, TTS), Document Intelligence (pre-built models), and key management across services.

AZURE CLOUD MASTERY

THANKS FOR WATCHING